and is being used by enterprises of all sizes. For a distributed database to be successful there are typical technology considerations like scale, consistency, resiliency, geo-replication and data locality. Given the growing popularity of multi-cloud, it's also critical for a distributed SQL database to be able to support multi-network deployment. YugabyteDB meets all these criteria.

Layered architecture

YugabyteDB has a layered architecture.

Yugabyte Query Layer (YQL) is the top layer that interfaces with applications interacting directly with it using client drivers. This layer manages API specific aspects for query/command compilation and runtime data representations. YQL is built with extensibility, allowing new APIs to be added as use cases grow.

Yugabyte SQL (YSQL) is a distributed SQL API built on top of the PostgreSQL language layer code. It's stateless SQL query engine is compatible with PostgreSQL.

Yugabyte Cloud QL (YCQL) is a semi-relational language that reflects Cassandra Query Language. Although it is SQL-like language, it's built specifically to be aware of the clustering of data across nodes.

DocDB is the distributed document store that has strong write consistency and is resilient to failures. It has automatic sharding, load balancing and geo aware data placement policies.

Sharding is done to tables stored in DocDB and is transparent to users.

Replication is applied with a configurable replication factor using the Raft consensus algorithm. It is performed at the table level, ensuring single row linearizability in case of failure.

Persistence is achieved with log-structured row/document-oriented storage including many optimisations supporting a growing data set for efficiency.

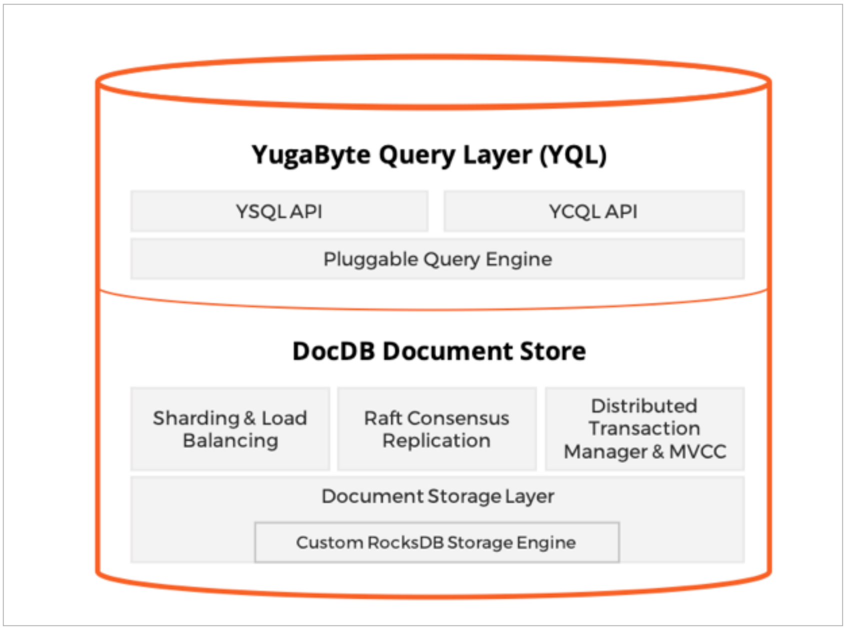

Figure 2: YugabyteDB architecture

Key features

What make YugabyteDB stand apart from other distributed databases are its features:

- Supports single- or multi-row transactions (OLTP apps)
- High performance and linear scalability
- Open source and cloud native with multi-cloud deployability
- Geo-location deployment and global data consistency
- Supports change data capture (CDC) mechanism
- Distributed and multi model
- Offers managed services

The need for data availability, consistency and optimum read/write performance is growing each day. YugaByteDB stands tall amongst all the distributed SQL databases. Its 5x storage sets it apart, given the growing cost of data storage, processing and analytics. END

References

[1] YugaByteDB, *https://www.yugabyte.com/*

[2] YugaByteDB Layered Architecture, *https://docs.yugabyte.com/preview/architecture/layered-architecture/*

[3] Forbes - Can Yugabyte become the de facto database for large scale, cloud native applications, *https://www.forbes.com/sites/brucerogers/2022/05/19/can-yugabyte-become-the-defacto-database-for-large-scale-cloud-native-applications/?sh=f4fc6a947885*

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